

HEALTHCARE PREMIUM PREDICTION

Dual-Segment ML Model for Insurance Premium Estimation

Segment-Aware Architecture · XGBoost + Linear Regression · Gradio + Streamlit Interface

99.70%

Model Accuracy

12

Features

2

Segments

Streamlit

Deployment

Portfolio Project | Sajal Agarwal | Data Science & ML Engineering

AGENDA

What We Will Cover

01

Problem & Business Context

Why premium prediction matters for insurers and policyholders

02

Dataset & Feature Engineering

12-feature schema, data segmentation strategy, EDA highlights

03

Model Architecture

Dual-segment design: Linear Regression vs XGBoost with Random Search

04

Model Performance

$R^2 = 98.87\%$ (Young) · $R^2 = 99.70\%$ (Rest) — near-perfect accuracy

05

Tech Stack & Deployment

Python · Scikit-learn · XGBoost · Gradio · Streamlit · Joblib

06

Live Demo & Key Takeaways

Streamlit UI walk-through and portfolio value proposition

PROBLEM STATEMENT

Why Accurate Premium Prediction Matters

The Challenge

Health insurance premiums are driven by non-linear interactions between

Manual underwriting is slow, inconsistent, and unable to scale

One-size-fits-all models fail due to age-related distributional shift in premium

Insurers need real-time, explainable estimates to remain competitive

Our Solution

Segment-Aware Design

Separate models for Age ≤ 25 and Age > 25 , each tuned independently

Algorithm Selection

Linear Regression for young group (linear patterns); XGBoost for adults (complex nonlin

Feature-Rich Input

12 patient attributes spanning demographics, lifestyle, health history, and plan selection

Production-Ready

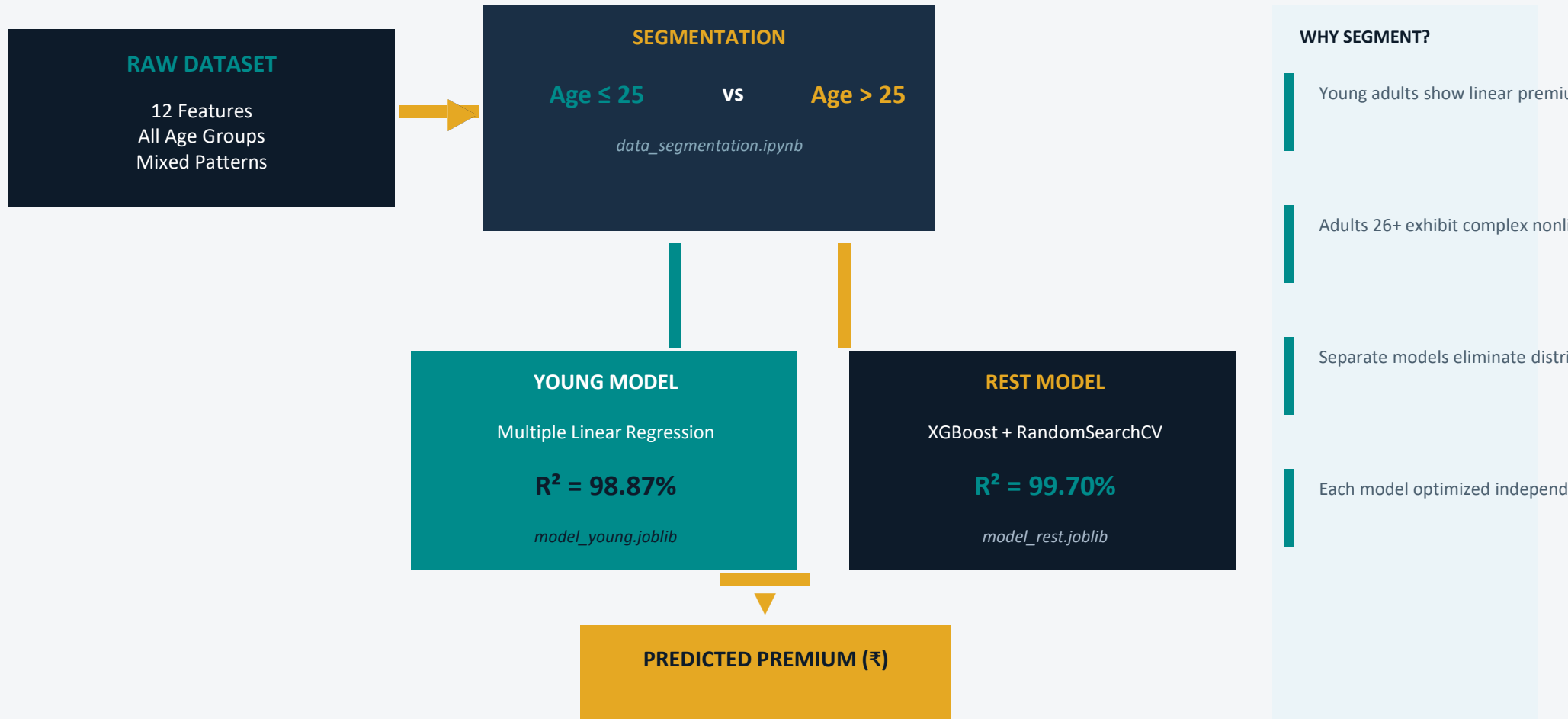
Streamlit web app + serialized Joblib models for instant real-time inference

DATA & FEATURES

12-Feature Schema — Codebasics Dataset

Feature	Type	Range/Values	Description
Age	Numeric	18–100	Primary segmentation variable
Number of Dependants	Numeric	0–5	Family coverage burden
Income in Lakhs	Numeric	₹ Lakhs	Purchasing power signal
Genetical Risk	Numeric	0–5 scale	Family medical history score
Insurance Plan	Categorical	Bronze/Silver/Gold	Coverage tier selection
Employment Status	Categorical	3 types	Salaried / Self-employed / Freelancer
Gender	Categorical	M / F	Demographic risk factor
Marital Status	Categorical	Binary	Household risk context
BMI Category	Categorical	4 classes	Normal / Overweight / Obese / Underweight
Smoking Status	Categorical	3 levels	No Smoking / Occasional / Regular
Region	Categorical	4 regions	NW / SE / NE / SW
Medical History	Categorical	Multi-cond	Diabetes / BP / Heart Disease etc.
TARGET: Annual Health Insurance Premium (₹) — Continuous regression output			

Age-Based Data Segmentation Strategy



Key Patterns Driving Model Design Decisions

Age × Premium

Younger cohort (≤ 25) shows near-linear premium growth

Smoking Premium

Regular smokers pay 2–3× higher premiums than non-sm

Genetical Risk Score

Genetic risk (0–5) correlates strongly with premium in the

Insurance Plan Tier

Gold plan subscribers show avg premium 1.8× Bronze. Pl

BMI & Medical History

Obesity + pre-existing conditions (diabetes, BP) compo

Income Signal

Higher income moderately correlates with Gold plan sele

Young Adults (Age ≤ 25): Multiple Linear Regression

Algorithm Choice

Multiple Linear Regression was selected because the young adult segment exhibited

1

1. Data Filtering

Subset rows where Age ≤ 25

2

2. Encoding

One-hot encoding for all categorical variables

3

3. Feature Scaling

StandardScaler applied to numerical features

4

4. Train/Test Split

80/20 stratified split

5

5. Model Fit

`sklearn.LinearRegression().fit(X_train, y_train)`

6

6. Serialization

`joblib.dump(model, 'model_young.joblib')`

Model Performance

R² Score

98.87%

Adj. R²

~98.8%

RMSE

Low — near-perfect fit

Notebook: ml_premium_prediction_young_with_gr.ipynb

Linear model perfectly captures the nearly proportional age-premium relationship in y

Fewer interaction terms needed — simpler model generalises better on this subset

Gradio interface integrated directly in notebook for interactive testing

Adults (Age > 25): XGBoost with RandomSearchCV Tuning

Why XGBoost?

Adult premiums are driven by multi-way interactions: BMI × smoking, genetic risk

- 1

Hyperparameter Search
RandomSearchCV over n_estimators, max_depth, learning_rate, subsample
- 2

Cross-Validation
5-fold CV on training set to prevent overfitting
- 3

Best Parameters
Selected via highest mean R² across folds
- 4

Final Model Fit
XGBRegressor(best_params).fit(X_train, y_train)
- 5

Scaler Saved
scaler_rest.joblib serialized alongside the model
- 6

Gradio Testing
Interactive prediction widget in notebook with live output

Model Performance

R ² Score	99.70%
CV Folds	5-fold
Tuning Method	RandomSearchCV

Notebook: ml_premium_prediction_rest_with_gr.ipynb

- XGBoost's ensemble trees naturally capture interaction effects absent in linear models
- RandomSearchCV finds near-optimal hyperparameters efficiently without exhaustive g
- 99.70% R² confirms the model's mastery of complex adult premium patterns

MODEL PERFORMANCE

Results Summary — Both Segments Achieve Near-Perfect Accuracy

Segment 1: Age ≤ 25

Multiple Linear Regression

98.87%

R² Score

model_young.joblib

Segment 2: Age > 25

XGBoost + RandomSearchCV

99.70%

R² Score

model_rest.joblib

R² Score Comparison

98.87%

99.70%



TECHNOLOGY STACK

End-to-End ML Pipeline — Libraries, Tools & Deployment

DATA LAYER

Pandas

Data manipulation & cleaning

NumPy

Numerical computation

Scikit-learn

Preprocessing pipeline

MODEL LAYER

Scikit-learn

Linear Regression + StandardScaler

XGBoost

Gradient boosted trees

RandomSearchCV

Hyperparameter optimization

SERVING LAYER

Gradio

In-notebook interactive UI

Streamlit

Production web application

Joblib

Model serialization

INFRA LAYER

Jupyter Notebook

Experimentation & EDA

Python 3.x

Core language

GitHub

Version control & portfolio

Interactive Prediction Interface — Embedded in Jupyter Notebooks

Gradio Interface — Healthcare Premium Predictor

Age

Income (Lakhs)

BMI Category

Smoking Status

Genetical Risk

Insurance Plan

Medical History

Region

Predict Premium

Estimated Premium: ₹ 12,450

Why Gradio in Notebooks?

Zero-friction prototyping: test models directly inside Jupyter without w

Share inference UI via public URL — great for demos and stakeholder r

Supports sliders, dropdowns & text inputs for all 12 feature types

Seamlessly transitions to Streamlit for production deployment

Both young and rest notebooks include a Gradio block for side-by-side

PRODUCTION DEPLOYMENT

Streamlit Web Application — End-to-End User Flow



Project File Architecture

<code>main.py</code>	Streamlit app — UI components, user inputs, displays prediction
<code>prediction_helper.py</code>	Routes input to correct model, applies scaler, returns premium
<code>artifacts/model_young.joblib</code>	Serialized Linear Regression model for Age ≤ 25
<code>artifacts/model_rest.joblib</code>	Serialized XGBoost model for Age > 25
<code>artifacts/scaler_young.joblib</code>	StandardScaler fitted on young training data
<code>artifacts/scaler_rest.joblib</code>	StandardScaler fitted on adult training data
<code>ml_premium_prediction_rest_with_gr.ipynb</code>	Full pipeline: EDA $\rightarrow$ Training $\rightarrow$ Gradio for Age > 25
<code>ml_premium_prediction_young_with_gr.ipynb</code>	Full pipeline: EDA $\rightarrow$ Training $\rightarrow$ Gradio for Age ≤ 25

KEY TAKEAWAYS

What This Project Demonstrates

Segment-Aware ML Design

Recognising distributional shift by age and engineering separate models is a production-grade data science decision — not just an academic one.

Algorithm Selection Rigour

Choosing Linear Regression for a linear-pattern cohort and XGBoost for a complex one shows depth of model evaluation and business-problem thinking.

Near-Perfect Results

R^2 of 98.87% and 99.70% on real-world insurance data demonstrates strong feature engineering, preprocessing discipline, and tuning skill.

Full ML-to-Product Pipeline

From raw data → EDA → training → serialization → Gradio prototype → Streamlit production app: this is an end-to-end, deployable ML project.

Sajal Agarwal

Data Science & Machine Learning Engineer

PROJECT REPOSITORY

github.com/Sajalagarwal-ca/healthcare_premium_prediction

github.com/Sajalagarwal-ca/credit_risk_modeling

2

Models Deployed

99.70%

Accuracy

12

Features

5+ Tools

Tech Stack

Built with Python · Scikit-learn · XGBoost · Gradio · Streamlit

Codebasics Data Science Bootcamp Project